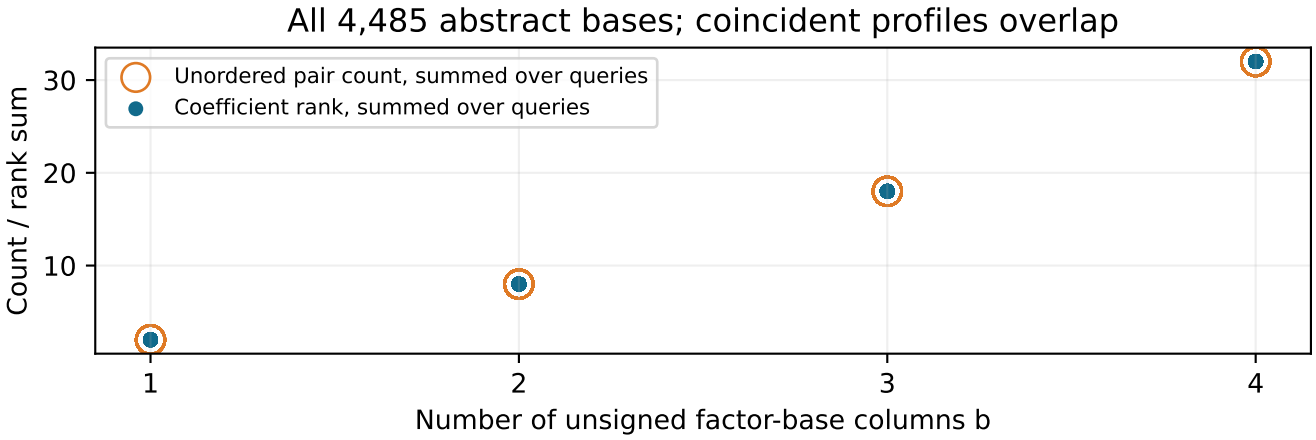
Signed two-summand rank

Finite census and a checked dependency mechanism

2026-10-29 PR2381 Native counts, not a speed result

Question. Does the complete signed two-summand S3 row model constrain empty-state useful rank? The lead atlas already records raw decomposition conservation (KN-FIND-007); row independence is a separate question. The original twist yield/degree/main-source-DLP objective remains open.

Exact finite population. Odd primes $r = 3, 5, 7, 11, 13, 17, 19, 23, 29, 31$; all $1 \le b \le \min(4, (r-1)/2)$ subsets of sign representatives; every nonzero query, complete two-summand rows, coefficients in $\mathbf{F}_r$. C++ ordered enumeration and independent Rust unordered construction agree on **4,485 bases,119,912 queries,zero rank/count defects**. These are abstract cyclic bases, not elliptic models.



All recorded bases plotted; overlapping profiles coincide. No sampling interval. Full rows are reconstructed independently and bound by exact stream hashes.

Both paths also match every row of the **96 frozen actual S3 arms /896queries** on the prior F43/r11 and F13/r7 source reductions. This checks old point-row bindings; it is not a new scalar-recovery experiment. All failures and policies remain in the frozen source records.

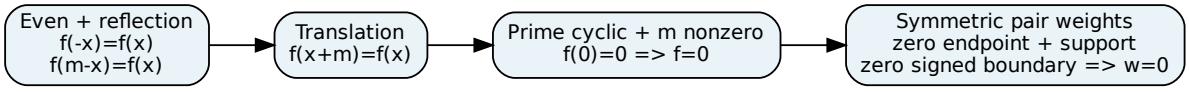
Domain control	Target	Rank	Pair count
$\mathbf{Z}/9$,base $\{1, 2, 4\}$,coefficients $\mathbf{F}_{101}$	3	2	3
$\mathbf{Z}/7$,base $\{1\}$,coefficients $\mathbf{F}_2$	2	0	1

The first removes target generation. The second loses the diagonal factor2. These retained losses prevent extending the finite observation without assumptions.

Validation. Both full native attempts pass;six changed/rehashed corruptions reject101/no receipt;22backend tests pass. The second attempt changes only the Lean-source binding after a simplification fix; all case/profile bytes and audit receipts match. No time,GAE,S,regularity or practical advantage claim.

Checked kernel, open matrix bridge

Five public named theorems /five audited declarations; zero failing or extra trusted axioms under the unmodified scoped repository proof gate. Only `propext,Classical.choice,Quot.sound`. Lean4.34.1/Mathlibd13f23b7.



For prime r,nonzero $m \in \mathbf{Z}/r$,an even function with $f(m-x) = f(x)$ and $f(0) = 0$ is zero: symmetry gives translation by$m$ and multiples of$m$ exhaust the group. For symmetric pair weights supported on $x + y = m$,zero at endpoint0,zero signed boundary therefore implies zero weights. These support and balance equations are written hypotheses.

Open formal binding. This is not yet a universal theorem that every signed coefficient-row family is independent. Diagonal weight conversion needs $2 \neq 0$; the general actual matrix-family binding and combinatorial count remain unformalized. The native census separately checks actual matrices in its bounds. Known raw counting would yield $2b^2/(r-1)$ mean rank *if* that complete independence binding holds. The18th source-bound IDC1 record checks only $(2b)(2b+1) - 2b = 4b^2$,32+16native points,with false-identity/source controls; it certifies neither independence nor speed.

Honest flow. Protocol/source were committed before main data. A declared one-call pilot precedes canonical JSON freeze; no threshold/population changed. Canonical protocol hash begins`a89bfebc5982a78e`;checker from atlas8b8800af. Two full native contracts completed,zero excluded/interrupted/resumed; repeated outputs are not independent samples. First actual Lean build failed on $f(0) = w(0, m-0)$ simplification; same theorem fixed and final gate passes. The original failed source/log remains,including its unaccepted sorry axiom. Export validation caught a pilot-only CLI guard added after the analysis hash. Exact frozen analyzer bytes were restored and both output sets replayed;the frozen protocol/hash was not rewritten. Missing JSON serializer and absent timing utility are separate infrastructure failures. No frozen prior source was patched.

What remains. A twist advantage in relevant degree/complete work, stationary useful yield,actual-state probability,broader geometry and full calibration remain open. Root5140-case library validation is still incomplete at600sec/124; naming/1921missing study registrations,required CI,coordination and integration remain held. No registry,benchmark or leaderboard row changed. No new speedup,universal impossibility,hardness,GHS or invalid-curve claim.

Reproduce. See adjacent README for exact protocol,source files,native runner,scoped Lean gate,byte-part case restoration and raw restoration. Native caps600sec;bounded iteration23:23–00:23UTC;two localworkers,no paidcompute. Approved branch checkpoint access works;credentials are outside Git/evidence. The next step must bind a useful actual matrix/probability result or a new relevant-degree encoding,while preserving the original main-source DLP goal.

Sources: frozen uniform/S3 and CM8/dual reports;Mathlib ZMod;lead-atlas KN-FIND-007,relation-graph-rank and Semaev-yield leads. Full source and retained failures linked in README. All numbers above copy native/audit records.